

Operation systems& Adv. Mobile project

Homework 1

Simple MyShell

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Remaining days 'till deadline: 2 days.

Introduction

This project implements a simplified shell program named goronoSH.

The goal of this program is to simulate the basic behavior of a Unix shell.

The main features include:

- Starting a custom shell from an executable file.
- Handling a prompt in the format `user@cwd$`.
- Accepting and executing commands available in the system's PATH.
- Executing external programs using `fork()` and `execvp()`.
- Waiting for child processes before reading the next command.
- A Makefile to compile, install, and remove the program.
- Supporting built-in commands:
 - *cd* (change directory).
 - *quit* (exit from goronoSH).
 - *history* (simple command history)

Building the program

Requirements:

- Operating System: Linux (Ubuntu, Debian, Fedora, Arch, etc.);
- Compiler: gcc or any C compiler compatible with POSIX;
- No external libraries are required.

Compilation with Makefile

The provided Makefile automates compilation and installation.

To build the program, run:

make

This creates an executable file named: *goronoSH*.

The program also can be locally installed in the current working directory with following command:

make install

Result: *goronoSH* will be copied into the *./bin* folder in current directory.

To uninstall the program run the following command:

make uninstall

It removes the installed program from the *./bin* folder in current directory.

To remove compiled object files and the executable run:

make clean

Usage Instructions

Run the shell: *./goronoSH*;

The prompt will look like: *user@/currentUser/currentPath\$*

You can run any command available in your system's PATH, for example:

ls

date

echo Hello

ping

traceroute

Built-in commands:

- *cd <path>* — change the working directory.
- *quit* — exit the shell.
- *history* — latest used commands.

To exit from the shell, run: *quit*.

Example of usage

```
gorono@assam:~/HW1$ ./grnSH
gorono@goronoSH:/home/gorono/HW1$ pwd
/home/gorono/HW1
gorono@goronoSH:/home/gorono/HW1$ ls -l
total 48
drwxrwxr-x 2 gorono gorono 4096  9월 28 17:50 bin
-rwxrwxr-x 1 gorono gorono 16992  9월 28 17:32 grnSH
-rw-rw-r-- 1 gorono gorono 2678  9월 28 17:32 hwl.c
-rw-rw-r-- 1 gorono gorono 12864  9월 28 17:44 hwl.o
-rw-rw-r-- 1 gorono gorono  442  9월 28 17:49 Makefile
gorono@goronoSH:/home/gorono/HW1$ cd ../
gorono@goronoSH:/home/gorono$ mkdir newdir
gorono@goronoSH:/home/gorono$ ls -l
total 48
-rw-r--r-- 1 gorono gorono 8980  9월  4 10:28 examples.desktop
-rw-rw-r-- 1 gorono gorono  518  9월 18 11:30 fork2.c
-rw-rw-r-- 1 gorono gorono 1841  9월 22 09:27 fork3.c
-rw-rw-r-- 1 gorono gorono  578  9월 18 10:34 fork.c
drwxrwxr-x 3 gorono gorono 4096  9월 28 18:21 HW1
drwxrwxr-x 2 gorono gorono 4096  9월 28 18:33 newdir
-rw-rw-r-- 1 gorono gorono  277  9월 18 10:22 pid.c
drwxrwxr-x 2 gorono gorono 4096  9월 23 12:10 PO1.1
drwxrwxr-x 5 gorono gorono 4096  9월 25 13:07 PO1.2
-rw-rw-r-- 1 gorono gorono  92  9월  8 10:12 prog.c
gorono@goronoSH:/home/gorono$ history
1  pwd
2  ls
3  cd
4  mkdir
5  ls
gorono@goronoSH:/home/gorono$ ifconfig -s
Iface      MTU      RX-OK RX-ERR RX-DRP RX-OVR      TX-OK TX-ERR TX-DRP TX-OVR Flg
docker0    1500      0      0      0  0           0      0      0      0  0 BMU
enp5s0f0    1500 21096781      0      1  0       3076445      0      0      0  0 BMRU
enp5s0f1    1500      0      0      0  0           0      0      0      0  0 BMU
lo          65536 1128944      0      0  0       1128944      0      0      0  0 LRU
gorono@goronoSH:/home/gorono$ quit
Exiting goronoSH...
gorono@assam:~/HW1$
```

Personal ideas and thoughts

Among the possible improvements I see:

1. the implementation of pipelines;
2. redirection of I/O streams;
3. input navigation using arrow keys;
4. autocomplete.